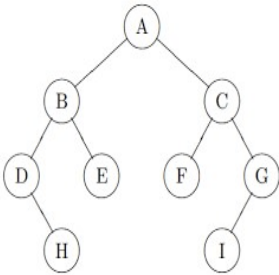
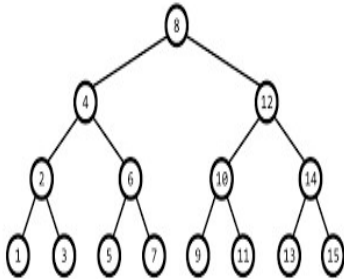
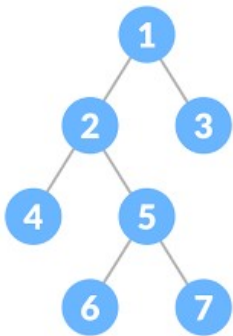
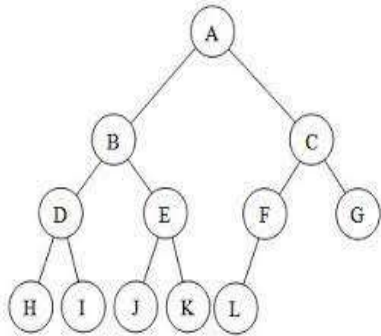


Summary of types of Binary tree with Formulas:

Terms	Binary Tree /2-Tree	Perfect Binary Tree	Full Binary Tree/ Proper Binary Tree/ Strictly Binary Tree	Complete Binary Tree
Possible no of tree	No of tree= $(2*N)!/(N+1)!N!$	-----	-----	-----
Rule	Maximum number of children are 2.	Insertion is not possible without changing the height All nodes have either 2 or 0 child Fill all level or all leaf nodes are on same level	All nodes have either 2 or zero child It is possible to insert nodes without increasing height (first condition must be followed after insertion as well) No one has one child	All nodes have degree of 0, 1, or 2 Complete level except last one. Always filled from left side only.
Diagram				
Finding Depth at Last Level	---	---	---	Depth of the last level = Number of digits in Binary Number (of given Nodes) -1

Maximum number of nodes of tree	$N_{\max} = 2^{(h+1)}-1$, height $h= 5$ So $N_{\max}=63$			
Minimum number of nodes	$N_{\min} =h+1$ height $h= 5$, $N_{\min}=6$	$N_{\min} = 2^{(h+1)}-1$ height $h= 5$, $N_{\min}=63$	$N_{\min} = 2h + 1$ height $h= 5$, $N_{\min}=11$	$N_{\min} = 2^h$ height $h= 5$, $N_{\min}=32$
Maximum possible height	$h_{\max} = N-1$ $N= 63$, $h_{\max} =62$	$h_{\max} = (\log(N+1)/\log 2) - 1$ $N= 63$, $h_{\max} =5$	$h_{\max} = (N-1)/2$ $N= 63$, $h_{\max} =5$	$h_{\max} = \log(N)/\log 2$ $N= 63$, $h_{\max} =5.97=5$
Minimum possible height	$h_{\min} = (\log(N+1)/\log 2) - 1$, No of Nodes= 63 so $h_{\min}=5$			
Maximum Nodes at level L	$N_{\max} = 2^L$, At level 5 maximum number of nodes 32			
Relation between leaf nodes and internal nodes	For Binary Tree			
	Total no of nodes = Number of internal nodes*degree of node +1			
	Ex. No of internal node with one degree are 5 and two degree are 10			
	Total no of nodes are $=(5*1)+(10*2)+1$, Ans:26			
	Number of leaf nodes = Number of total nodes - Number of internal nodes			
	Ex. Total number of nodes are 26-Internal nodes(10+5), Ans:26-15=11			
	For Full Binary Tree			
	Total number of nodes= $(2*\text{no of leaves})-1$ Ex. No of leaves nodes=10, Ans:19			
Array representation of binary tree				
Index of left child	$2i + 1$ (where $i =$ index start form 0) $2i$ (where $i =$ index start form 1)			
Index of right child	$2i + 2$ (where $i =$ index start form 0) $2i + 1$ (where $i =$ index start form 1)			
Index of Parent node of given child	$i-1/2$ (where $i =$ index start form 0) $i/2$ (where $i =$ index start form 1)			